CSD 416 PROJECT PHASE II PRESENTATION

STEGDETECT

GROUP - 011

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Project Guide
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Abstract

- **Hide Secret Messages Safely**
  Users can hide sensitive text or files inside images and audio with encryption for added security.

- **Detect Hidden Information**
  The system analyzes media files using statistical methods to find concealed data accurately.

- **Fast and Practical Web Platform**
  Built with React and FastAPI, it offers smooth performance, API support, and real-world usability.

Background and Literature Review

Existing Systems:

1. **SilentEye (2011)**
   A lightweight, cross-platform steganography utility focused on usability and educational demonstration.

   - **Key Findings:**
     - GUI-driven steganography toolkit for image and audio media.
     - Supports **BMP, JPG, WAV** formats.
     - Implements basic LSB-based embedding for conceptual clarity.
     - Optimized for accessibility rather than resilience.
   - **Limitations:**
     - Static, non-adaptive embedding pipeline.
     - Low robustness against compression artifacts, signal noise, and statistical steganalysis.
     - No automation, payload optimization, or adaptive region selection.

Background and Literature Review

Existing Systems:

2. F5 Algorithm (2012)

A frequency-domain JPEG steganography technique leveraging matrix encoding to reduce embedding distortion.

- **Key Findings:**
  - Operates in the **DCT coefficient domain** rather than spatial pixels.
  - Uses **matrix encoding** to minimize modification rate.
  - Employs permutation and shrinkage to reduce detectability.
  - Improved imperceptibility over naïve LSB methods.
- **Limitations:**
  - JPEG-only; inherently lossy domain.
  - Embedding patterns statistically predictable under advanced steganalysis.
  - No adaptive learning or dynamic payload optimization.

Background and Literature Review

Existing Systems:

3. DeepSteg (2019)

A CNN-based end-to-end steganographic framework using deep neural networks for adaptive data hiding.

- **Key Findings:**
  - Encoder–decoder architecture trained via convolutional neural networks.
  - Learns data-driven embedding without handcrafted heuristics.
  - Achieves high reconstruction fidelity (PSNR > 36 dB).
  - Adaptive feature-space embedding.
- **Limitations:**
  - High computational and dataset requirements.
  - Limited interpretability of learned embedding regions.
  - Vulnerable to adversarial perturbations and distribution shift.

Gap Analysis

- Existing tools focus mainly on **data hiding**, not detection.
- Support is **limited to specific formats** (BMP, WAV, etc.)
- Many are **complex or not user-friendly**.
- Few tools are built for **education and hands-on learning**.
- **No single platform** combines hiding & detection in a simple interface.
- **StegDetect bridges this gap** with integrated encode, decode, and analysis for learners.

Problem Statement

To design and develop a user-friendly web-based application that enables secure data hiding using steganography techniques and simultaneously provides steganalysis features to detect hidden information in digital media.

Objectives

- Build a **secure web-based platform** for steganalysis and steganography.
- Allow users to **easily hide and extract data** from images or audio files.
- Develop **detection mechanisms** to identify hidden information.
- Ensure **strong data confidentiality** and resistance to attacks.
- Provide a **clean, intuitive interface** for seamless user experience.
- Combine **data hiding and detection** into one unified system.

Introduction & Motivation

- **Secure data transmission** is critical in modern digital communication systems. Traditional encryption protects data content but does not hide the existence of communication.
- Steganography conceals information within digital media such as **images, and audio.**
- Most existing tools **focus only on data hiding or detection** and lack unified functionality. They also offer limited media support or are difficult for beginners to use.
- **StegDetect** provides a unified, web-based platform for steganography and steganalysis. It supports learning, ethical usage, and research in cybersecurity and digital forensics.

Proposed Methodology

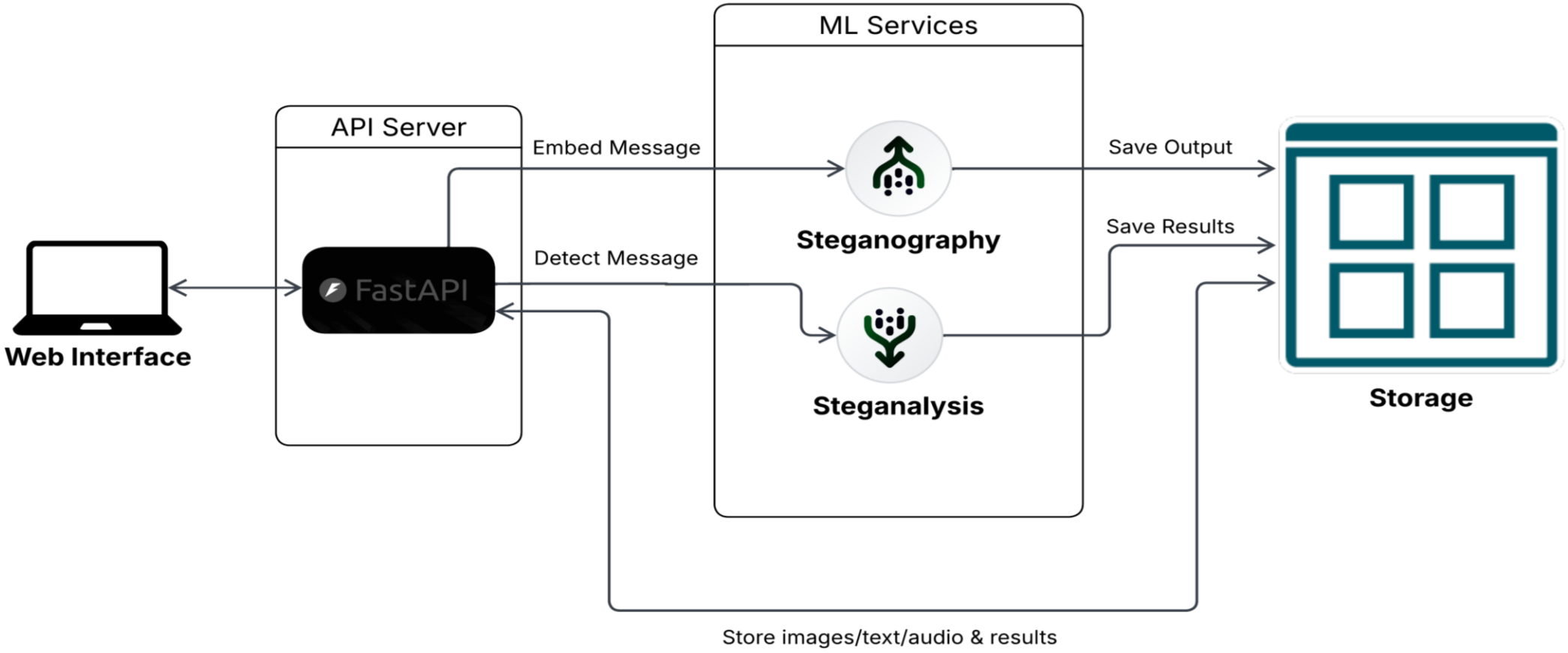
1. **Input:** Upload an image (PNG) or audio (WAV) file as cover media, along with your secret message.
2. **Preprocessing:** Compress the secret payload to save space and normalize the cover media. If a password is provided, securely encrypt the payload.
3. **Steganography:**
   - **Image:** Hide data quietly within the Least Significant Bits (LSB) of the image pixels.
   - **Audio:** Hide data within the Least Significant Bits (LSB) of the audio samples.
4. **Steganalysis:**
   - Extract the payload using length signatures and error checking.
   - Test images using statistical checks (Chi-Square & RS Analysis) to detect if someone has hidden secrets inside.
5. **Output:** Receive the modified stego-media or the safely recovered secret message.

Proposed Methodology

Technology Used:

- **Git, GitHub** – Version control
- **React using Vite, TypeScript, TailwindCSS, Shadcn (for UI components)** – Frontend & UI
- **FastAPI** – Backend APIs
- **Supabase (PostgreSQL , Auth, Storage)** – Database & Login
- **OpenCV, Pillow, NumPy** – Processing & ML
- **Docker** – Deployment & Setup (Optional)

System Architecture

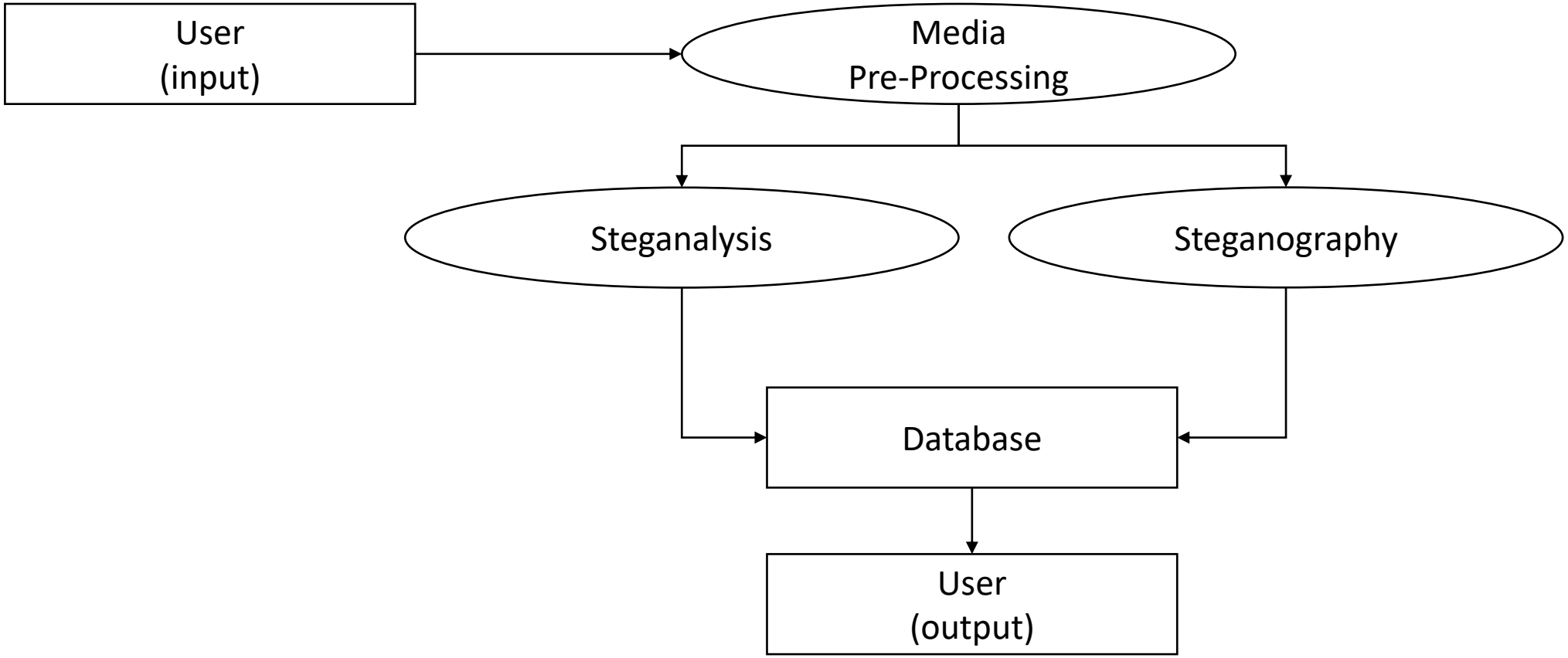


Data Flow Diagram

1.DFD LEVEL 0

Data Flow Diagram

2. DFD LEVEL 1



Hardware & Software Requirements

Hardware Requirements

- **Processor:** Dual-core CPU (Intel i3 / AMD Ryzen 3 or higher)
- **RAM:** 8 GB (16 GB recommended for ML tasks)
- **Storage:** 20 GB free disk space

Software Requirements

- **Operating System**: Windows 10 / Ubuntu 22.04+ / macOS (latest stable)
- **Development Tools**: Git (Version Control, Node.js (frontend), Python 3.10+ (backend)
- **Backend & Database**: Supabase (PostgreSQL + Auth + Storage)
- **Libraries**: OpenCV, Pillow, NumPy
- **Deployment (Optional)**: Docker

Works Completed

Development Summary:

- ✓ **UI/UX Design Completed:** User interfaces for image and audio steganography modules are fully designed and implemented.
- ✓ **Modular Components :** Core React components are completed.
- ✓ **Data Integration Implemented:** Mock and real backend APIs are integrated to enable smooth frontend–backend communication.
- ✓ **Enhanced User Experience:** Dynamic loading states, validation messages, and interactive feedback are implemented for better usability.
- ✓ **Backend Completed:** Backend development is fully completed, including API endpoints, media processing logic, steganography/steganalysis functions, authentication, and database integration.

Results & Demonstration

StegDETECT

Log in

PRODUCT FEBRUARY 22, 2026

Introducing StegDETECT

We've trained a highly advanced model called StegDETECT which identifies covert data hidden within digital media. The forensic engine makes it possible for security researchers to uncover steganographic payloads, analyze modified bit-planes, challenge malicious masking techniques, and extract hidden communications.

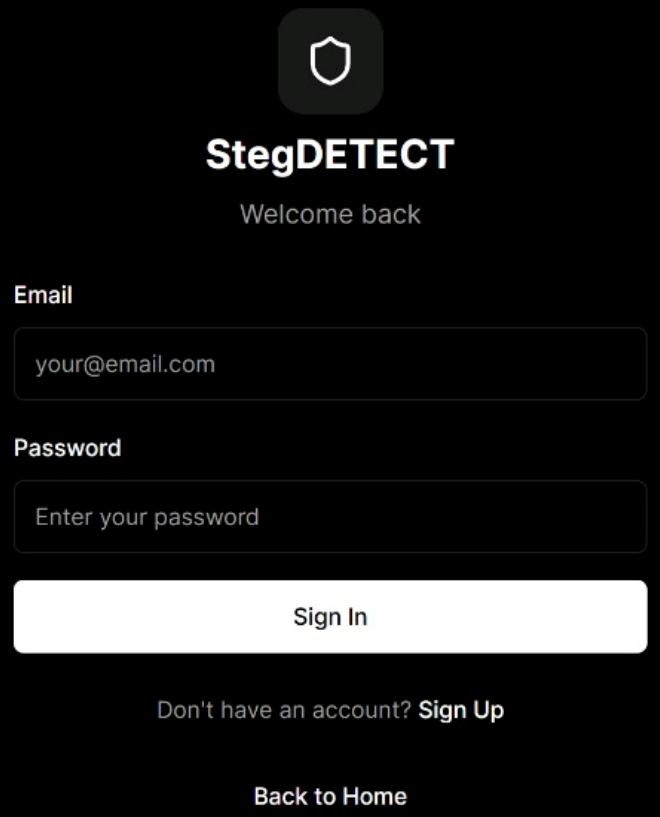
Try StegDETECT ↗

StegDETECT is a sibling model to our Vision-Forensic suite, which is trained to follow deep pixel-level analysis instructions and provide detailed reports on media integrity.

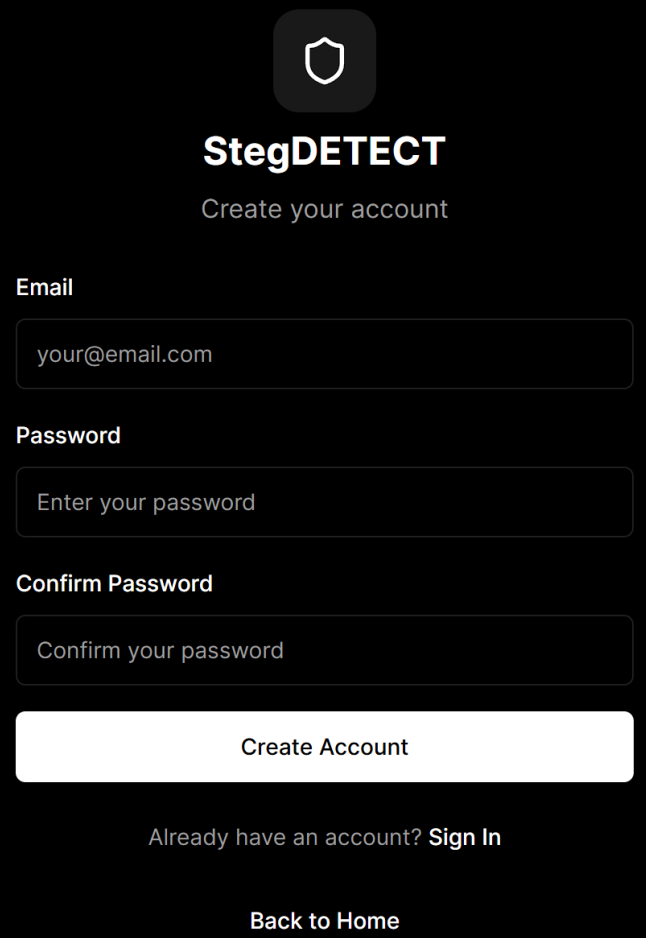
We are excited to introduce StegDETECT to get feedback from the cybersecurity community and learn about its capabilities against zero-day encoding methods. During the research preview, usage of the StegDETECT API is free.

Samples

Sign In



Sign Up



StegDETECT

Dashboard

- Steganography
- Steganalysis
- History
- Settings

Steganography

Hide messages, audio, or images inside image or audio

1 Carrier Media

Select the image or audio file that will hide the secret

File Upload

Drop your file here

or click to browse • Image, Audio • Max 300MB

✓ Supports PNG (images)
✓ Supports WAV (audio files)

2 Secret Payload

What do you want to hide?

Text Message | File (Audio/Image)

Message

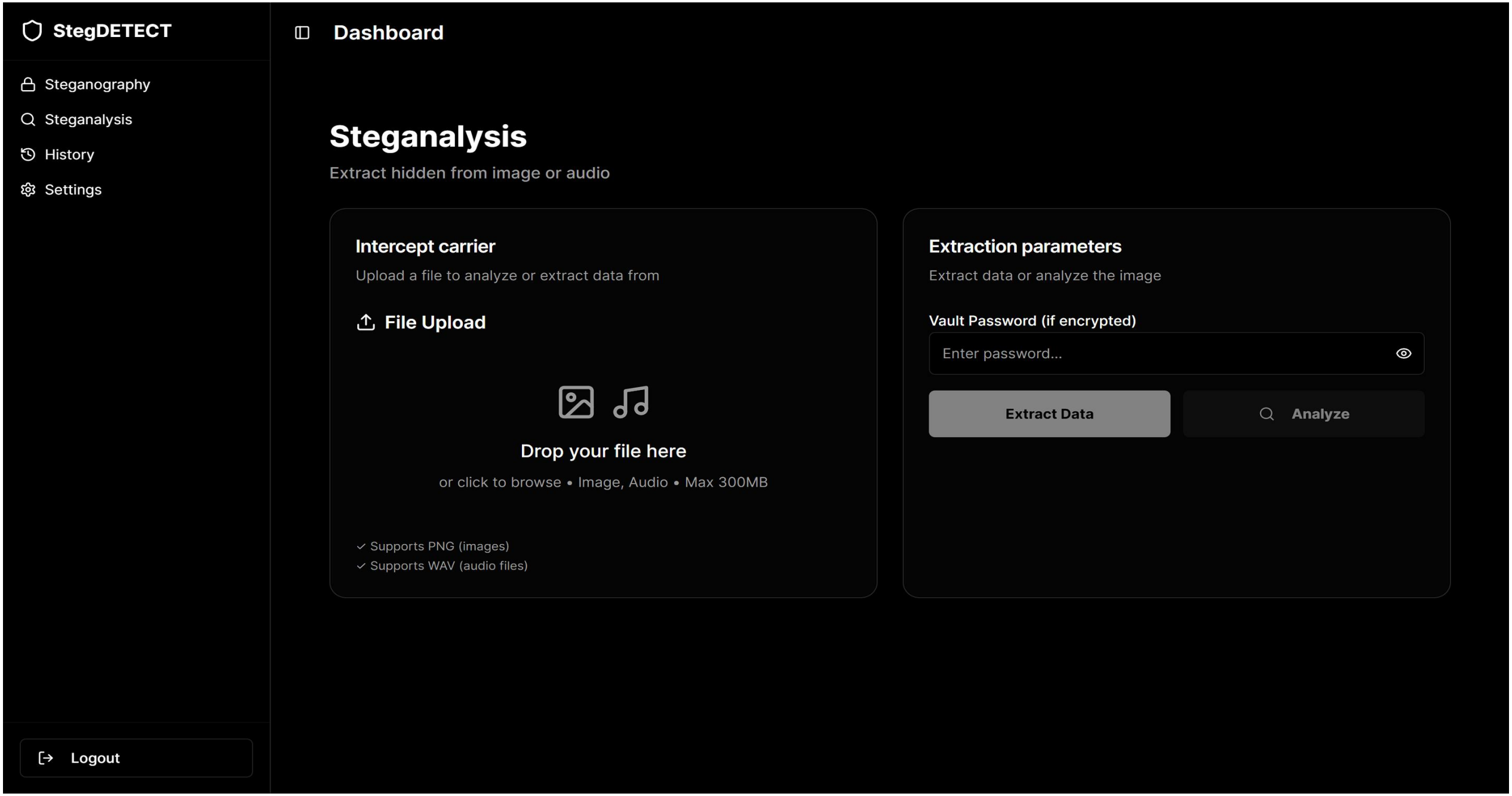
Enter secret text...

Security Password (Optional)

Encryption password...

Hide Secret Data

Logout


StegDETECT
Dashboard
Steganography
Steganalysis
History
Settings
Logout
Steganalysis
Extract hidden from image or audio
Intercept carrier
Upload a file to analyze or extract data from
File Upload
Drop your file here
or click to browse • Image, Audio • Max 300MB
✓ Supports PNG (images)
✓ Supports WAV (audio files)
Extraction parameters
Extract data or analyze the image
Vault Password (if encrypted)
Enter password...
Extract Data
Analyze

StegDETECT

Dashboard

Main Menu

- Steganography
- Steganalysis
- History
- Settings

Logout

History

View your recent steganography and analysis tasks

Search & Filter

Find specific tasks

Search by filename...

Recent Activity

Your steganography and steganalysis history

stego_pexels-njeromin-35999359.png 2026-02-24 04:28	Steganography	Success
stego_pexels-njeromin-35999359.png 2026-02-24 04:18	Extraction	Error
stego_pexels-njeromin-35999359.png 2026-02-23 11:38	Steganography	Success
stego_file_example_WAV_2MG.wav		

StegDETECT

Dashboard

- Steganography
- Steganalysis
- History
- Settings

Logout

Settings

Manage your account and preferences

Profile Settings

Update your personal information

Full Name

John Doe

Email

john@example.com

Save Profile

Preferences

Customize your experience

Email Notifications
Receive email updates about your tasks

Auto-save History
Automatically save all operations to history

Advanced Mode
Show advanced steganography options

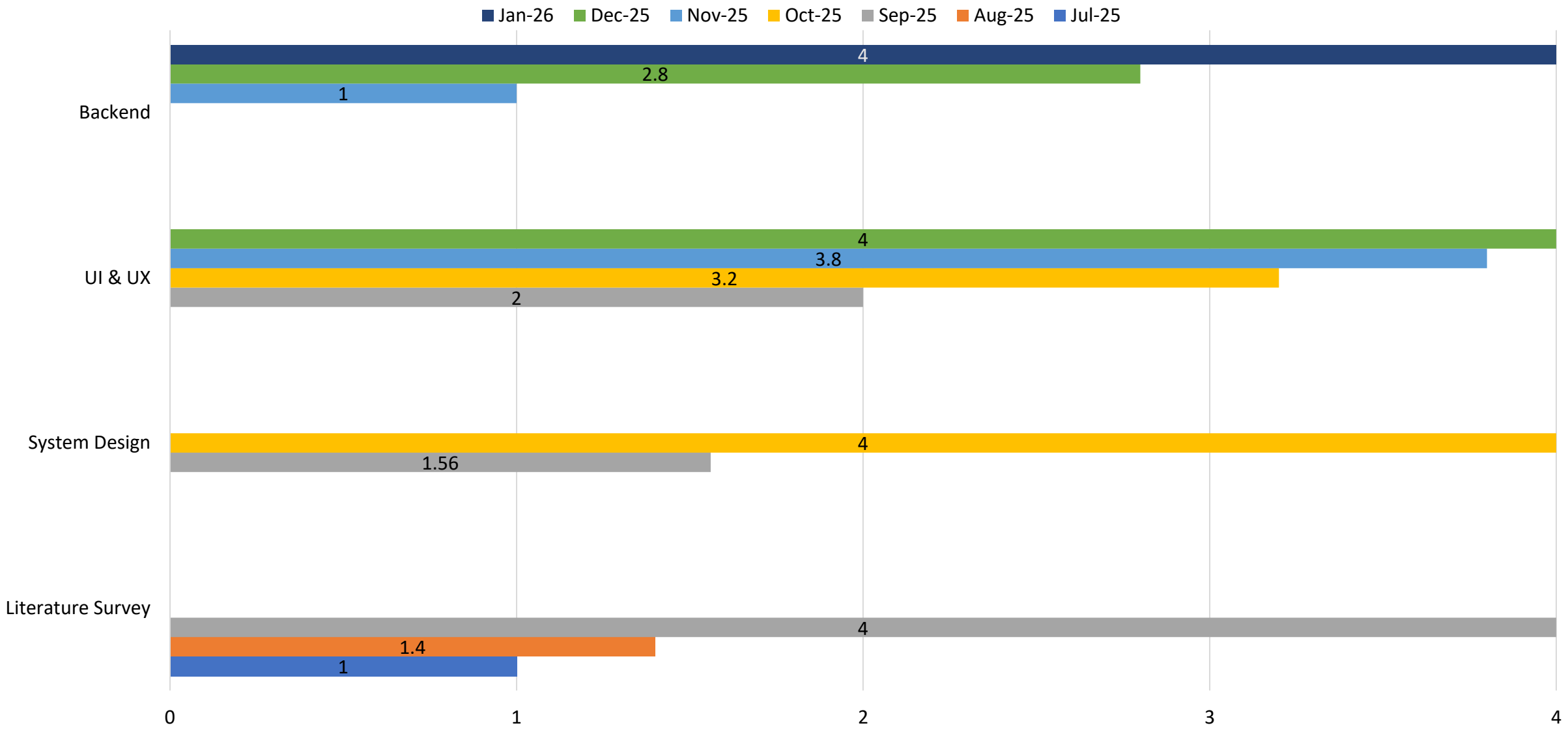
Key Results and Findings

- **Seamless Hiding:** Successfully embedded text payloads within image and audio cover files with zero visible or audible quality degradation.
- **Precision Extraction:** Achieved 100% accurate extraction and reconstruction of hidden messages from modified carrier files.
- **Proactive Detection:** Effectively identified altered stego-media using advanced mathematical noise modeling and statistical analysis.
- **Intuitive Experience:** Delivered a user-friendly, unified dashboard that simplifies uploading, encoding, decoding, and steganalysis.
- **Robust Encryption:** Ensured payload security by integrating AES-GCM password protection, rendering extracted data unreadable without authorization.

Learning Outcomes

- **Steganography & Steganalysis**: Mastered the techniques for hiding data seamlessly and running statistical detection against carrier files.
- **Teamwork & Collaboration**: Gained practical experience coordinating and working as a team to deliver on shared project goals.
- **Frontend Development**: Built a responsive, dynamic web application from scratch using modern React.js.
- **Cybersecurity Knowledge**: Deepened our understanding of covert communication channels, digital forensics, and payload security.
- **User-Centric Design**: Successfully designed and delivered an intuitive, user-friendly dashboard for complex tools.

Project Timeline



Conclusion

- **Educational:** Provides a hands-on learning tool for students and researchers to understand steganography, digital forensics, and cybersecurity.
- **Research:** Serves as a foundational platform for prototyping and testing new steganography and steganalysis algorithms.
- **Awareness:** Highlights the security risks of hidden data, training professionals to identify covert channels and secure data.
- **Digital Forensics:** Acts as a basic utility for investigators to perform initial scans on suspicious files.

Future Work

- Advanced **AI and deep learning–based steganalysis** can be integrated to improve detection accuracy against sophisticated hiding techniques.
- Support for **additional media formats** such as video and compressed files can be added to extend the system's applicability.
- **Real-time detection and monitoring** features can be implemented for large-scale network and forensic investigations.
- Enhanced **security features and cloud scalability** can be introduced to support high-volume users and enterprise-level deployments.

Future Impacts

- **Advanced Data Privacy:** Securely hide sensitive information inside images or audio to prevent unauthorized access.
- **Stronger Cybersecurity:** Detect hidden malware and covert data leaks using advanced steganalysis.
- **Digital Copyright Protection:** Embed invisible watermarks to safeguard ownership and prevent piracy.
- **Secure Communication:** Enable safe information sharing in high-surveillance or censored environments.

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Thank You…